

DETAILED DESCRIPTION OF THE INVENTION — MANUFACTURING METHODS AND FABRICATION VARIATIONS

The intraluminal physiologic monitoring device may be manufactured using a variety of fabrication processes selected to achieve desired mechanical properties, biocompatibility, cost efficiency, scalability, and sensor integration capability.

In various embodiments, device bodies may be produced through injection molding, compression molding, liquid silicone rubber molding, transfer molding, extrusion processes, or multi-shot molding techniques allowing integration of regions having differing material properties within a single structure.

Additive manufacturing methods may also be employed, including stereolithography, selective laser sintering, fused filament fabrication, digital light processing, or other three-dimensional printing techniques capable of producing complex geometries, internal channels, embedded structures, or rapid prototype iterations.

Flexible electronic assemblies may be fabricated using printed flexible circuits, stretchable conductive traces, thin-film deposition, screen printing of conductive inks, laser-patterned substrates, or laminated multilayer electronics integration. Sensors may be embedded during molding operations or integrated post-fabrication.

Overmolding techniques may encapsulate electronic assemblies within compliant materials to create sealed biocompatible housings. Insert molding processes may allow structural components, electrodes, or reinforcement elements to be permanently incorporated within molded bodies.

Surface finishing operations may include polishing, plasma treatment, coating deposition, laser texturing, micro-patterning, or chemical surface modification designed to optimize comfort, durability, or sensing performance.

Assembly methods may utilize automated robotic assembly, ultrasonic welding, adhesive bonding, thermal welding, mechanical fastening, magnetic coupling, or hybrid joining techniques depending on device architecture.

Quality control procedures may include leak testing, electrical validation, dimensional verification, sterilization compatibility testing, or automated inspection systems configured to ensure reliable performance across production batches.

Manufacturing workflows may be adapted for high-volume consumer production, small-batch clinical manufacturing, or research-scale fabrication without departing from the principles disclosed herein.

The manufacturing methods described are illustrative rather than limiting. Any fabrication or assembly technique capable of producing an intraluminal physiologic monitoring device with substantially similar functional characteristics is considered within the scope of the present disclosure.